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Title: Noninvasive Ventilation in PICU & HD

The use of non-invasive support is becomingwidely accepted as first modality of treatment for respiratory distress forchildren.

Its use is associated with reduced need forintubation as well as reduced number of extubation failure.

Mechanismof action of NIV

* Maintain lung volume during expiration

*Decrease atelectasis and Improve oxygenation

*Unload respiratory muscleand reduce fatigue

Indicationsof use of NIV:

· Respiratorydistress secondary to bronchiolitis , pneumonia or congestive heart failure

· Respiratorysupport post extubation

· Weaningfrom BIPAP support

· Respiratorysupport to children with neuromuscular disease

· Postextubation strider

**Pleasenote that for a baby with apnea on non invasive support, careful monitoringshould be done in highly monitored area (PICU). You need to consider intubationin cases where the baby is unsafe (having frequent apnea, need totransfer from area to another).

Pre-requests for successful NIV

· *Appropriateinterface (face mask , nasal prongs , nasal masks)

· *Adequatesedation

One can use oral medications like chloralhydrate, oral clonidine. Other options include dexmedetomidine infusion andthis is to be used in PICU and require close monitoring for heart rate andlevel of sedation.

High flow nasal cannulae treatment (HFNC)

This isdefined as delivery of blended, heated, humidified air and oxygen through nasalcannulae at flow rates higher than 2 L/min.

Flowrate can range from 2L/min to 40L/min depending on maximum flow rates availablein the blender. The aim is to provide continuous distending pressure (PEEP) tounload respiratory muscles and improve work of breathing.

Contraindicationsto HFNC:

- Choanalatresia
- Recentsurgery to nasopharynex , repair for tracheoesophgeal fistula
- Recentabdominal surgery

Starting HFNCO:

- Childneed to be on continuous monitoring for respiratory rate, heart rate , Spo2 andwork of breathing
- Start HFNC at 1l/kg/min , for example for a child weight 4 kg , you can start at 4l/min
- Start Fio2 that keep the child saturation >92%. You can set FIO2 0.4 to startwith.
- Maximum flow is 2l/kg/min and is decided on maximum flow in blender.
- Obtain BGA or ABG 30 minutes post commencing HFNC , ensure appropriate blood gasreading

**Please notethat child may have normal oxygen saturation despite hypercarbic respiratoryfailure

- Monitorfor features of improvement , the heart rate and respiratory rate as well aswork of breathing should improve with lung recruitment
- Atthe end of 2 hours from commencing HFNC , decide if the treatment is working ornot. Observe improvement in heart rate , respiratory rate and work of breathing
- Considerchanging form of non invasive support or mechanical ventilation if there is noclinical improvement. The decision need to be taken with the consultant forPICU or HD.

** for childrenadmitted in high dependency area , the child need to be transferred to PICU ifescalation of respiratory support is needed.

Please notethat we sometimes use other forms of non invasive support in HD, which will bedecided by HD consultant.

- Allchildren need to have nasogastric feeding tubes which need to be aspiratedevery 2-4 hours for air.

Complications of HFNC:

- Gastricdistention
- Pneumothorax(rare)

Neonatal CPAP, bi-level CPAP devices

Infantflow SIPAP (fusion care)

Infantflow SIPAP provide non invasive support for small babies and infants throughthe use of small nasal masks or prongs.

Thefollowing modes are available on SIPAP devices:

· Startwith PEEP of 6-7 cm and adjust Fio2 according to the baby oxygen saturation.

· Ensurethat there is no significant leak in the system and that the baby get a properlevel of PEEP

Other modes on SIPAP

Biphasic: time triggered pressure assists. Inthis mode, we can set both desired PEEP and PS

Neonatal bubble CPAP

Aim of bubble CPAP is to provide continuouspositive airway pressure to allow lung recruitment

Setting bubble CPAP

· Adjustthe desired PEEP level according to clinical decision .Suggested starting PEEP6-7 cm H2O

· Thebaby needs oro-gastric feeding tube for air decompression. Air should beaspirated every 2 -4 hours

· makesure that the chamber is bubbling all the times and that there is no leak inthe system , this ensure positive pressure is delivered

· ensurethat there is no disconnection in the system at the time of care (avoid lossof PEEP)

· obtainBGA or ABG 30 minutes post commencing bubble CPAP

· considerintubation in cases the baby did not show signs of improvement after 2 hours(i.e there is no signs of improvement in the work of breathing , presence ofrespiratory acidosis or Fio2 > 50 % or PEEP> 8 with no clinicalimprovement)

Non invasive devices for older children(Bi-pap)

The provision of BI-PAP support includesproviding EPAP and IPAP. This can be achieved using servo I ventilator orTRIOLOGY.

IPAP (inspiratory positive airway pressure) isused to assist spontaneous breaths (PS). On the other hand, EPAP (expiratorypositive airway pressure) is used to provide constant PEEP to keep alveoli openand stent airway.

Setting BIPAP:

- - startwith PEEP of 8 and PS of 10.Higher levels of PEEP of 10-12 can be considered inolder children

- - setFIo2 0.4 to keep oxygen saturation > 92%

- - observefor respiratory distress , heart rate and respiratory rate

- - obtainBGA or ABG 30 minutes from commencing the treatment

- - lookfor evidence of improvement (improvement in respiratory rate , heart rate andoxygen saturation)

Contraindications to BIPAP support

· * Recentfacial , upper airway surgery

· * Presenceof facial abnormalities (burns , trauma)

· *Presenceof vomiting

· * Recentupper GIT surgery

· * Presenceof copious respiratory secretions

Predictors of non invasive ventilation failure

The following factors are associated with noninvasive ventilation failure

· *needto change forms of non invasive support

· *Presenceof respiratory acidosis with PH 7.25 after 2 hours from treatment

· *Underlyingmulti organ dysfunction

Indications of mechanical ventilation

- - Failure of non invasive support
- - Respiratoryarrest
- - Cardiorespiratoryarrest
- - Inabilityto protect the airway

Weaning from non invasive support

Weaning from non invasive support should beconsidered once the child clinical condition improves. Weaning can start after1st day of non invasive ventilation use.

If the child is on high flow nasal cannulatreatment, wean the flow gradually until the flow < 1l/kg/min where one canconsider changing the child to nasal cannula . Otherwise, PEEP can be graduallyreduced until it is minimum PEEP (newborn: 5 cm H2O, child 6 cm H2O).

In select groups of patients, consideration canbe made to change them from CPAP / BIPAP to HFNC.

References:

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
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